# Rohan Wagle

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## Education

**Ashoka University** | BA (Hons. with Research) in Economics 2027
Minor in Mathematics · 3.51/4 CGPA

## Skills

R, Python, Stata, SQL, Excel, LaTeX, GitHub

## Internships and Experience

**Ashoka University and DAI Advisory** | Research Assistant Apr 2026 – Jul 2026

- Co-managed daily field operations and survey logistics for a large-scale randomized controlled trial (RCT) tracking a baseline universe of over 7,000 respondents across target markets
- Diagnosed live delivery risks such as respondent refusals, postponements, and eligibility shifts for a 69-person field team (57 enumerators, 12 supervisors), re-sequencing deployment daily and testing targeted pitch strategies to keep the sample on track

**Purnartha Investment Advisers Pvt. Ltd.** | Equity Research Intern May 2025 – Jul 2025

- Standardized and validated financial data for 300+ listed Indian companies, designing repeatable cleaning and cross-checking routines that caught inconsistencies before they propagated into valuation models and sector reports
- Reframed an open-ended research question — *which Indian firms move together, and why?* — as a measurable graph problem, building a quantitative framework in Python (NetworkX) to model inter-firm linkages (supply-chain, ownership, industry) and estimate performance spillovers

## Publications

**A Bayesian Approach to Partial Homogeneity in Staggered Difference-in-Difference** (with Parush Arora) Jul 2026
Ashoka University Economics Discussion Paper Series

- Designed and ran the Monte Carlo framework in R benchmarking a novel $\ell_0$-penalized estimator against standard Two-Way Fixed Effects across 11 simulated data-generating environments
- Quantified the bias–variance tradeoff, proving a 26–52% reduction in estimator variance with no attenuation bias, and automated 40+ diagnostic dashboards to keep every run auditable
- Stress-tested the estimator on Cengiz et al. (2019)'s minimum-wage study, detecting heterogeneous treatment effects that the pooled specification masked
- Developed and published `phdid`, an R package on CRAN implementing the paper's estimators, heterogeneity tests, and sampler diagnostics

## Projects

**Efficient PCA Implementation on Sparse Data Matrices** Jan 2026 – Apr 2026
Data Science, Multivariate Statistics

- Implemented the Block Power Method for PCA from scratch in R with a custom implicit-centering algorithm, eliminating the memory blow-up from matrix densification that makes off-the-shelf implementations fail on sparse data
- Recovered a low-rank subspace on MovieLens 100K capturing 35.28% of variance, then built gender-stratified loading plots to segment the user base, applying dimensionality reduction techniques for customer segmentation and targeting

**Predicting Secondary School Student Performance** Jul 2025 – Aug 2025

Machine Learning and Data Analysis

- Built a supervised ML pipeline in R predicting student outcomes from demographic, social and academic features, replicating Cortez & Silva (2008) as an explicit benchmark before attempting to beat it
- Improved predictive accuracy 15% over the published model through feature engineering and hyperparameter tuning, evaluating each iteration against the benchmark rather than tuning to a single metric

**Independent Research Study** | Advised by Prof. Anuradha Saha Jan 2026 – Apr 2026

Macroeconomics

- Modelled how AI-induced task restructuring, labour search frictions, and endogenous skill acquisition interact in developing economies, simulating firm training incentives against worker skill choices across three equilibria
- Identified an "AI learning gap" showing how premature AI adoption traps economies in low-skill equilibria, and converted it into actionable policy diagnostics for regulatory standards

## Positions Held

**Editor** | Ashoka University Economics Review Sep 2025 – Apr 2026

- Reviewed 5+ student-authored empirical papers for the university's flagship peer-reviewed journal, evaluating their identification strategy, robustness and replicability and returning structured, actionable feedback to authors
- Scaled interest in quantitative research across the student body by running coding workshops and maintaining shared digital repositories for research roles and conferences

**Junior Editor** | The Edict Sep 2023 – Apr 2026

- Authored and edited 10+ long-form pieces for an audience of 1,000+, structuring primary interviews and field evidence into clear long-form articles for a non-specialist readership

**Peer Tutor** | Undergraduate Writing Program, Ashoka University Aug 2025 – Dec 2025

- Served as sole teaching assistant for a class of 15 first-year students in Introduction to Critical Thinking, the university's compulsory foundation course
- Guided class discussions and ran a writing workshop, reviewing essay drafts and returning feedback in weekly one-on-one office hours

## Languages Known

English, Hindi, and Marathi (Native)